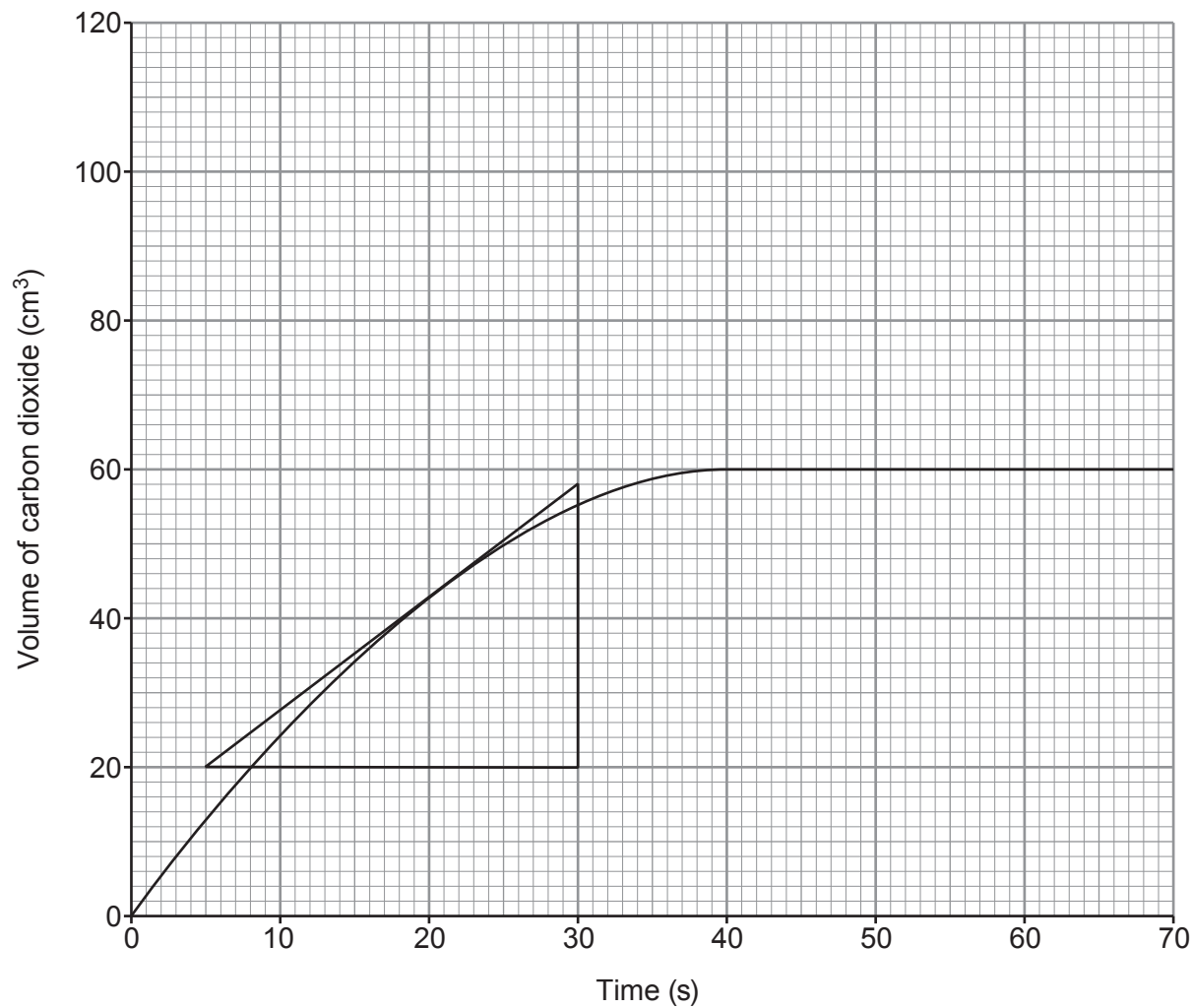


11. The graph shows the volume of gas produced in a reaction between 0.40 g of calcium carbonate powder and excess 1 mol/dm³ hydrochloric acid.



(a) Use the tangent drawn to calculate the rate of the reaction at 20 s. [2]

Rate = cm³/s



Examiner
only

- (b) (i) On the same grid, sketch the graph that would be seen if the reaction were repeated with 0.60 g of calcium carbonate powder and acid of the same concentration and still in excess. [2]
- (ii) Explain the graph that you have sketched using your knowledge of particle theory. [2]

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6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2

Group

3

4

5

6

7

0

1 H Hydrogen 1										4 He Helium 2																									
7	Li	9	Be											11	B	12	C	14	N	16	O	19	F	20	Ne										
	Lithium		Beryllium												Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon										
	3		4												5		6		7		8		9		10										
23	Na	24	Mg											27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar										
	Sodium		Magnesium												Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon										
	11		12												13		14		15		16		17		18										
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Potassium		Calcium		Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
	19		20		21		22		23		24		25		26		27		28		29		30		31		32		33		34		35		36
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	103	Rh	106	Pd	108	Ag	112	Cd	115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
	Rubidium		Strontium		Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Rhodium		Palladium		Silver		Cadmium		Indium		Tin		Antimony		Tellurium		Iodine		Xenon
	37		38		39		40		41		42		43		44		45		46		47		48		49		50		51		52		53		54
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	192	Ir	195	Pt	197	Au	201	Hg	204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
	Caesium		Barium		Lanthanum		Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Gold		Mercury		Thallium		Lead		Bismuth		Polonium		Astatine		Radon
	55		56		57		72		73		74		75		76		77		78		79		80		81		82		83		84		85		86
223	Fr	226	Ra	227	Ac																														
	Francium		Radium		Actinium																														
	87		88		89																														

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410UA0-1



S23-3410UA0-1

FRIDAY, 16 JUNE 2023 – MORNING

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and
Essential Resources
HIGHER TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in Question 6(a).

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	11	
3.	9	
4.	8	
5.	5	
6.	9	
7.	12	
8.	9	
9.	8	
Total	80	

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